JX Control Families Book

End-to-end coverage of Control, Image, Theme, Run, Drawing, Path, Output, and native executable proof contracts.

Direction: append future chapters at the end and append the same chapter to the running table of contents.

Running Table Of Contents

Append rule: add new family chapters after the last chapter, then add one new TOC row. Existing meanings should get modification notes instead of silent rewrites.

Chapter 1: Family Map

Chapter Preface

JX controls are host-neutral contracts, not browser-only widgets. The family map keeps the language appendable as it grows toward native windows, X11 relevance, and a JX window server.

Chapter Terminology Appendix

Term	Meaning
family	Top-level taxonomy owner for a contract object.
Control	Host controls and drawing operations.
Image	Image leaves, replacement sets, repeated path images, and trade-offs.
Theme	Spin, zoom, and mash movement composition.
Run	Percent-window behavior such as pong and reset.
Output	Callback connector with a movement property bag.

Inner Functionality

Every contract is designed to survive renderer changes. A browser can render it today, a Win32 host can render it now, and a later JX window server can own it without changing the Book model.

Future chapters should append to the end of this book and add one line to the running table of contents.

Placebo Code

```
control image.any uses Image, Theme, Run, and Output
host renders it as browser, Win32, X11, or JX window server
```

Tight Version

```
Control::image('image.any', 'Any image type', $src, 'image/*', [
    'pin' => Control::imagePin(Control::XY_CENTER, Control::XY_LB, Control::XY_RT),
]);
```

Advanced Techniques

Advanced technique: keep semantic families stable, then allow each host to optimize rendering locally. The same `data-control` JSON can become HTML attributes, Win32 draw calls, or future compositor messages.

Modifications And Meanings

This chapter is append-friendly. Add new functions as new rows in the terminology appendix, then add placebo and tight examples below the existing examples.

Chapter 2: Control Family

Chapter Preface

Control is the surface family. It gives Books a way to declare inputs, switches, image controls, and drawing surfaces without binding the Book to a particular OS toolkit.

Chapter Terminology Appendix

Term	Meaning
text	Normal form input.
spin	Numeric spinner or dial-like stepper.
toggle	Boolean switch control.
drawing	Host-renderable drawing surface.
image	MIME/source-backed image control.
pin	A point or setting that anchors a control to motion or host layout.

Inner Functionality

A control has `id`, `type`, `props`, and `events`. The HTML renderer exposes the contract through `data-control`; native hosts can read the PHP `contract()` array directly.

Controls can carry image replacement sets and themes. That allows a spin control to be a dial image in one host and a native number spinner in another.

Placebo Code

```
make a spin control
pin it
give it a dial image set and a spin theme
```

Tight Version

```
$spinTheme = Theme::spinClicks('spin.rate', 1, 2, 12, ['wrap' => true]);

Control::spin('spin.rate', 'Spin control', 3, [
    'min' => -12,
    'max' => 12,
    'step' => 1,
    'pin' => true,
    'theme' => $spinTheme,
]);
```

Advanced Techniques

Advanced technique: correlate a visible control with an event-sourced output bag. The control remains a contract while the host chooses whether to render native input, an image replacement, or a themed custom surface.

Modifications And Meanings

This chapter is append-friendly. Add new functions as new rows in the terminology appendix, then add placebo and tight examples below the existing examples.

Chapter 3: Image Family

Chapter Preface

Image is a full family because controls need replaceable, repeatable, and event-sourced image behavior. A neon trail is not a special line type; it is a line carrying an image brush.

Chapter Terminology Appendix

Term	Meaning
img	One image leaf.
IMG_DOTTED	Repeat the same image one after another along a path.
IMG_BLUR	Repeat every x pixels for a blur or speed trail.
replacementSet	State-indexed images for dials, buttons, and switches.
tradeOff	Event record describing one image changing to another.
ROLE_DIAL	Replacement-set role for dial controls.
ROLE_BUTTON	Replacement-set role for button controls.
ROLE_SWITCH	Replacement-set role for switch controls.

Inner Functionality

Image controls accept any MIME-backed image source. SVG is supported but not special.

Replacement images let a control trade visual state without changing the control identity. Trade-offs should be consumed as events, not inferred from markup.

Placebo Code

```
use neon-line.png as a repeated paint image
when image.view changes, trade button-up for button-disabled
```

Tight Version

```
Image::blur('neon-line.png', 'image/png', 8, [
    'role' => 'paint',
    'glow' => 0.9,
]);

Image::tradeOff(
    'evt-image-view-toggle',
    'control.image.view.changed',
    Image::img('controls/button-up.png', 'image/png'),
    Image::img('controls/button-disabled.png', 'image/png'),
    'View display switched off',
);
```

Advanced Techniques

Advanced technique: image replacement sets can be shared by controls and executable hosts. The compiled Windows proof includes dial, switch, and button replacement sets in JSON.

Modifications And Meanings

This chapter is append-friendly. Add new functions as new rows in the terminology appendix, then add placebo and tight examples below the existing examples.

Chapter 4: Theme Family

Chapter Preface

Theme owns motion behavior that can be reused across controls, lines, and curves. This lets the host mash spin and zoom into one movement without losing the inner contracts.

Chapter Terminology Appendix

Term	Meaning
spinClicks	Maps a degree span to a number of clicks.
zoom	Maps one scale to another with easing.
mash	Combines several theme motions into one behavior.
clicksPerDegree	Derived ratio used for click and dial handling.
wrap	Host hint that spin can wrap through its range.

Inner Functionality

`Theme::mash([$spinTheme, $zoomTheme])` has no extra name or mode. The kind is `mash`; the motions array holds the inner behavior.

The themed executable uses this to spin, move, and zoom a dial-like image control along a curve.

Placebo Code

```
between degree 1 and degree 2, consume 12 clicks
while moving, zoom from 1.0 to 1.35
mash both motions together
```

Tight Version

```
$spinTheme = Theme::spinClicks('spin.rate', 1, 2, 12, ['wrap' => true]);
$zoomTheme = Theme::zoom(1.0, 1.35, 'ease-out');
$mashTheme = Theme::mash([$spinTheme, $zoomTheme]);
```

Advanced Techniques

Advanced technique: when a host receives pointer input, it can convert path phase into spin clicks, zoom scale, and image replacement state in one callback cycle.

Modifications And Meanings

This chapter is append-friendly. Add new functions as new rows in the terminology appendix, then add placebo and tight examples below the existing examples.

Chapter 5: Drawing And Path Family

Chapter Preface

Drawing is the shape side of controls. It supports simple shapes, curved shapes, movement curves, and SVG path evocation for host-native path APIs.

Chapter Terminology Appendix

Term	Meaning
line	Straight movement or drawing line.
curve	Movement path defined by degree/control points.
polygon	Closed straight-edged shape.
curvedShape	Closed curved shape from points.
path	SVG-style path evoker.
smooth	Interpolation property from 0 to 1.
path evoker	A preserved path string the host maps to its own path API.

Inner Functionality

`curve()` is for movement. `path()` is for path drawing or path evocation. Hosts may use the same graphics backend for both, but the contract keeps their meanings separate.

The current HTML renderer emits SVG. The native executable uses a WinForms graphics path to prove the same direction outside the browser.

Placebo Code

```
draw a polygon
draw a curved shape
evocate an SVG-style path
move a control along a curve
```

Tight Version

```
Control::polygon('triangle-shape', [
    ['x' => 258, 'y' => 24],
    ['x' => 334, 'y' => 76],
    ['x' => 280, 'y' => 124],
]);

Control::path(
    'svg-evoker',
    'M 40 42 C 76 6 118 74 156 42 S 238 78 316 42',
    ['fill' => 'none', 'stroke' => '#be185d'],
);
```

Advanced Techniques

Advanced technique: click-to-seek maps pointer location to the nearest curve phase, then updates movement, spin, zoom, trail, and callback output.

Modifications And Meanings

This chapter is append-friendly. Add new functions as new rows in the terminology appendix, then add placebo and tight examples below the existing examples.

Chapter 6: Run Family

Chapter Preface

Run describes how a movement proceeds over a percent window. It is the difference between bouncing back and jumping back.

Chapter Terminology Appendix

Term	Meaning
pong	Move forward and then backward.
reset	Move forward and jump back to start.
start	Percent where movement begins.
end	Percent where movement completes.
percent window	The active part of a full path run.

Inner Functionality

Default run window is 0..1. A run window of 0.25..0.75 uses only the middle half of the path.

Legacy boolean `true` still normalizes to `Control::pong(0, 1)`, but new code should use explicit run constructors.

Placebo Code

```
run the middle half of the line
when reaching 75 percent, jump back to 25 percent
```

Tight Version

```
Control::line(
    'reset-line',
    ['x' => 42, 'y' => 166],
    ['x' => 318, 'y' => 166],
    Control::reset(0.25, 0.75),
);
```

Advanced Techniques

Advanced technique: percent windows let a host reuse one full geometry while multiple controls occupy different spans of it.

Modifications And Meanings

This chapter is append-friendly. Add new functions as new rows in the terminology appendix, then add placebo and tight examples below the existing examples.

Chapter 7: Output Callback Family

Chapter Preface

Movement can produce output where the user picked it. The output connector names the callback and carries the property bag.

Chapter Terminology Appendix

Term	Meaning
output	Callback connector on a movement.
callback	Host event or function name to call.
at	Screen anchor for output placement.
bag	Movement property payload.
picked	User or host-selected movement point.

Inner Functionality

When a user clicks a path, the host can call the output connector with position, phase, run window, and the supplied bag.

The themed executable shows this as a small callback bag panel near the picked point.

Placebo Code

```
when reset-line is picked
show output at XY_RT
call controls.movementPicked with the movement bag
```

Tight Version

```
Control::line(
    'reset-line',
    ['x' => 42, 'y' => 166],
    ['x' => 318, 'y' => 166],
    Control::reset(0.25, 0.75),
) + [
    'output' => Control::output(
        'controls.movementPicked',
        Control::XY_RT,
        ['source' => 'reset-line', 'run' => 'reset'],
    ),
];
```

Advanced Techniques

Advanced technique: output callbacks make controls feel native because the output appears where the action happened, not in a detached log.

Modifications And Meanings

This chapter is append-friendly. Add new functions as new rows in the terminology appendix, then add placebo and tight examples below the existing examples.

Chapter 8: Executable Proofs

Chapter Preface

The executable proofs show that the language direction is not only server-rendered markup. The contracts can be compiled into native Windows executables.

Chapter Terminology Appendix

Term	Meaning
jx-spec-contract.exe	Compiled JSON contract proof.
jx-themed-window.exe	Native WinForms visual proof.
click-to-seek	Clicking near the path moves control to that phase.
callback bag panel	Visible output of picked movement properties.

Inner Functionality

`jx-spec-contract.exe` emits JSON with Control, Image, Theme, Run, Drawing, Path, and Output sections.

`jx-themed-window.exe` draws a native window with mashed motion, click-to-seek, blur/dotted trails, a reset runner, and output bags.

Placebo Code

```
build native proof
show mashed image control
click path
emit property bag where picked
```

Tight Version

```
tools\windows\build-spec-contract.ps1
build\windows\jx-spec-contract.exe --smoke
build\windows\jx-spec-contract.exe --contract

tools\windows\build-themed-window.ps1
build\windows\jx-themed-window.exe
```

Advanced Techniques

Advanced technique: the executable can be treated as a temporary native host while JX grows toward a larger window-server architecture.

Modifications And Meanings

This chapter is append-friendly. Add new functions as new rows in the terminology appendix, then add placebo and tight examples below the existing examples.

Appendix A: Current Term Tally

Term	Meaning
Families	Control, Image, Theme, Run, Output
Control terms	text, spin, toggle, drawing, image, pin, imagePin, paintPoint, imageDisplay, polygon, curvedShape, path, curve, line
Image terms	img, IMG_DOTTED, IMG_BLUR, replacementSet, tradeOff, ROLE_DIAL, ROLE_BUTTON, ROLE_SWITCH
Theme terms	spinClicks, zoom, mash, clicksPerDegree
Run terms	pong, reset, start, end, percent window
Output terms	output, callback, at, bag, picked
Executable terms	jx-spec-contract.exe, jx-themed-window.exe, click-to-seek, callback bag panel

Appendix maintenance rule: whenever a new term appears in a chapter, add it to this tally. This keeps the language family map visible as the book grows.